

Nishat Raihan

Provost's Postdoctoral Fellow, University of Notre Dame

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SUMMARY

Postdoctoral Fellow with deep expertise in the end-to-end development of **Large Language Models**, from data curation and pre-training to fine-tuning and evaluation. Proven track record of releasing open-source tools that improve model performance in complex, low-resource, and multilingual settings. Experienced instructor and teaching assistant, with a record of teaching and supporting CS courses (including C/C++ and programming/algorithms), leading labs and recitations, and grading at scale. **I do not require H1B sponsorship.**

ACADEMIC

Provost's Postdoctoral Fellow

University of Notre Dame

Aug. 2026 – Aug. 2028

Notre Dame, IN, USA

- Faculty mentor: Dr. Joanna C. S. Santos

Ph.D. in Computer Science

George Mason University

Aug. 2021 – July 2026

Fairfax, VA, USA

- Dissertation: *Adapting Code LLMs for Underrepresented Domains*
- Advisors: Dr. Marcos Zampieri and Dr. Antonios Anastasopoulos
- CGPA: **3.96/4.00** (Distinguished Academic Achievement Award)

M.S. in Computer Science

George Mason University

Aug. 2021 – Dec. 2024

Fairfax, VA, USA

- CGPA: **3.97/4.00** (Distinguished Academic Achievement Award)

B.Sc. in Computer Science and Engineering

Islamic University of Technology

Jan. 2015 – Nov. 2018

Gazipur, Bangladesh

- CGPA: **3.70/4.00** (ranked 7th of 51)

HONORS AND AWARDS

Provost's Postdoctoral Fellowship, College of Engineering, University of Notre Dame

2026–2028

Doctoral Research Scholarship, Tier 1, The Graduate Division, George Mason University

2026–2027

Distinguished Academic Achievement Award (Ph.D.), George Mason University (GPA > 3.90)

2026

Distinguished Academic Achievement Award (M.S.), George Mason University (GPA > 3.90)

2024

Best Paper Award, 1st Workshop on Bangla Language Processing (BLP @ EMNLP)

2023

BIBLIOMETRICS

40+ Publications

561+ Citations

h-index 13

i10-index 18

PUBLICATIONS

Author name in **bold**. Mentee co-authors marked with a superscript ^m (graduate-student mentees).

Journal Articles

- J2** **Nishat Raihan**, Dhiman Goswami, Sadiya Sayara Chowdhury Puspo, Mohammed Latif Siddiq, Christian Newman, Tharindu Ranasinghe, Joanna C. S. Santos, Marcos Zampieri. “On the Performance of Large Language Models on Introductory Programming Assignments.” *Journal of Intelligent Information Systems*, Springer, 2025. [PDF]
- J1** Rafsanjany Kushol, Imamul Ahsan, **Md. Nishat Raihan**. “An Android-Based Useful Text Extraction Framework Using Image and Natural Language Processing.” *International Journal of Computer Theory and Engineering (IJCTE)*, vol. 10, no. 3, pp. 77–83, 2018. [PDF]

Conference Papers

- C12** **Nishat Raihan**, Antonios Anastasopoulos, Marcos Zampieri. “TigerCoder: A Novel Suite of LLMs for Code Generation in Bangla.” *Proceedings of the Language Resources and Evaluation Conference (LREC 2026)*.
- C11** **Nishat Raihan**, Noah Erdachew, Jayoti Devi^m, Joanna C. S. Santos, Marcos Zampieri. “CodeGuard: Improving LLM Guardrails in CS Education.” *Findings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2026)*. [PDF]
- C10** **Nishat Raihan**, Marcos Zampieri. “TigerLLM: A Family of Bangla Large Language Models.” *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)*, Short Papers. [PDF]
- C9** **Nishat Raihan**, Antonios Anastasopoulos, Marcos Zampieri. “mHumanEval: A Multilingual Benchmark to Evaluate Large Language Models for Code Generation.” *Proceedings of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL 2025)*, Main, Long Papers. [PDF]
- C8** **Nishat Raihan**, Joanna C. S. Santos, Marcos Zampieri. “MojoBench: Language Modeling and Benchmarks for Mojo.” *Findings of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL 2025)*. [PDF]
- C7** **Nishat Raihan**, Mohammed Latif Siddiq, Joanna C. S. Santos, Marcos Zampieri. “Large Language Models in Computer Science Education: A Systematic Literature Review.” *Proceedings of the 56th ACM Technical Symposium on Computer Science Education (SIGCSE-TS 2025)*. [PDF]
- C6** **Nishat Raihan**, Christian Newman, Marcos Zampieri. “Code LLMs: A Taxonomy-based Survey.” *2024 IEEE International Conference on Big Data (IEEE BigData 2024)*. [PDF]
- C5** **Nishat Raihan**, Dhiman Goswami, Sadiya Sayara Chowdhury Puspo, Christian Newman, Tharindu Ranasinghe, Marcos Zampieri. “CSEPrompts: A Benchmark of Introductory Computer Science Prompts.” *27th International Symposium on Methodologies for Intelligent Systems (ISMIS 2024)*. [PDF]
- C4** **Md Nishat Raihan**, Sadiya Sayara Chowdhury Puspo^m, Shafkat Farabi, Ana-Maria Bucur, Tharindu Ranasinghe, Marcos Zampieri. “MentalHelp: A Multi-Task Dataset for Mental Health in Social Media.” *2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*. [PDF]
- C3** **M. Raihan**, N. Ulfat, N. Saqib. “An Experimental Analysis on the Sensitivity of the Most Widely Used Edge Detection Methods to Different Noise Types.” *International Conference on Computing Advancements (ICCA 2022)*. [PDF]
- C2** T. Ahmed, A. Rahman, T. M. Chowdhury, R. Kushol, **M. N. Raihan**. “A Novel Approach to Classify Electrocardiogram Signals Using Deep Neural Networks.” *2nd International Conference on Computer and Information Sciences (ICCIS 2020)*, IEEE. [PDF]
- C1** T. Ahmed, **M. N. Raihan**, R. Kushol, M. S. Salekin. “A Complete Bangla Optical Character Recognition System: An Effective Approach.” *22nd International Conference on Computer and Information Technology (ICCIT 2019)*, IEEE. [PDF]

Workshop and Shared-Task Papers

- W18** **Nishat Raihan**, Marcos Zampieri. “Temporal Text Classification with Large Language Models.” *Proceedings of the Workshop on Natural Language Processing for Digital Humanities (NLP4DH @ ACL 2026)*.
- W17** **Nishat Raihan**, Sadiya Sayara Chowdhury Puspo, Ana-Maria Bucur, Stevie Chancellor, Marcos Zampieri. “Large Language Models for Mental Health: A Multilingual Evaluation.” *2nd Workshop on Language Models for Low-Resource Languages (LoResLM @ EACL 2026)*. [PDF]
- W16** **Nishat Raihan**, Mohammad Anas Jawad, Md Mezbaour Rahman, Noshin Ulfat, Pranav Gupta, Mehrab Mustafy Rahman, Shubhra Kanti Karmakar, Marcos Zampieri. “Overview of BLP-2025 Task 2: Code Generation in Bangla.” *Proceedings of the Second Workshop on Bangla Language Processing (BLP @ IJCNLP-AACL 2025)*. [PDF]

- W15** Wren McQueary, Sadia Afrin Mim, **Md. Nishat Raihan**, Justin Smith, Brittany Johnson. “Py-holmes: Causal Testing for Deep Neural Networks in Python.” *Companion Proceedings of the 32nd ACM International Conference on the Foundations of Software Engineering (FSE 2024)*. [PDF]
- W14** Matthew Shardlow et al. (incl. **M. N. Raihan**). “The BEA 2024 Shared Task on the Multilingual Lexical Simplification Pipeline.” *19th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2024)*. [PDF]
- W13** Matthew Shardlow et al. (incl. **M. N. Raihan**). “An Extensible Massively Multilingual Lexical Simplification Pipeline Dataset Using the MultiLS Framework.” *3rd Workshop on Tools and Resources for People with Reading Difficulties (READI @ LREC-COLING 2024)*. [PDF]
- W12** **Nishat Raihan** et al. “EmoMix-3L: A Code-Mixed Dataset for Bangla-English-Hindi Emotion Detection.” *WILDRE Workshop, LREC-COLING 2024*. [PDF]
- W11** **Md Nishat Raihan**, Dhiman Goswami, Al Nahian Bin Emran, Sadiya Sayara Chowdhury Puspo, Amrita Ganguly, Marcos Zampieri. “MasonTigers at SemEval-2024 Task 9: Solving Puzzles with an Ensemble of Chain-of-Thoughts.” *18th International Workshop on Semantic Evaluation (SemEval @ NAACL 2024)*. [PDF]
- W10** D. Goswami, S. S. C. Puspo, **M. N. Raihan**, A. N. B. Emran, A. Ganguly, M. Zampieri. “MasonTigers at SemEval-2024 Task 1: An Ensemble Approach for Semantic Textual Relatedness.” *18th International Workshop on Semantic Evaluation (SemEval @ NAACL 2024)*. [PDF]
- W9** S. S. C. Puspo, **M. N. Raihan**, D. Goswami, A. N. B. Emran, A. Ganguly, O. Uzuner. “MasonTigers at SemEval-2024 Task 8: Performance Analysis of Transformer-based Models on Machine-Generated Text Detection.” *18th International Workshop on Semantic Evaluation (SemEval @ NAACL 2024)*. [PDF]
- W8** A. Ganguly, S. S. C. Puspo, D. Goswami, **M. N. Raihan**. “MasonPerplexity at ClimateActivism 2024: Integrating Advanced Ensemble Techniques and Data Augmentation for Climate Activism Stance and Hate Event Identification.” *LT-EDI Workshop, EACL 2024*. [PDF]
- W7** A. Ganguly, A. N. B. Emran, S. S. C. Puspo, **M. N. Raihan**, D. Goswami, M. Zampieri. “MasonPerplexity at Multimodal Hate Speech Event Detection 2024: Hate Speech and Target Detection Using Transformer Ensembles.” *7th Workshop on Challenges and Applications of Automated Extraction of Socio-political Events from Text (CASE @ EACL 2024)*. [PDF]
- W6** D. Goswami, S. S. C. Puspo, **M. N. Raihan**, A. N. B. Emran. “MasonTigers at LT-EDI-2024: An Ensemble Approach towards Detecting Homophobia and Transphobia in Social Media Comments.” *4th Workshop on Language Technology for Equality, Diversity, Inclusion (LT-EDI @ EACL 2024)*. [PDF]
- W5** **[Best Paper] Md Nishat Raihan**, Umma Hani Tanmoy, Anika Binte Islam, Kai North, Tharindu Ranasinghe, Antonios Anastasopoulos, Marcos Zampieri. “Offensive Language Identification in Transliterated and Code-Mixed Bangla.” *1st International Workshop on Bangla Language Processing (BLP @ EMNLP 2023)*, Singapore. [PDF] [Dataset]
- W4** **Md Nishat Raihan**, Dhiman Goswami, Sadiya Sayara Chowdhury Puspo, Marcos Zampieri. “nlpBDpatriots at BLP-2023 Task 1: A Two-Step Classification for Violence Inciting Text Detection in Bangla.” *1st International Workshop on Bangla Language Processing (BLP @ EMNLP 2023)*, Singapore. [PDF]
- W3** Dhiman Goswami, **Md Nishat Raihan**, Sadiya Sayara Chowdhury Puspo, Marcos Zampieri. “nlpBDpatriots at BLP-2023 Task 2: A Transfer Learning Approach to Bangla Sentiment Analysis.” *1st International Workshop on Bangla Language Processing (BLP @ EMNLP 2023)*, Singapore. [PDF]
- W2** Dhiman Goswami, **Md Nishat Raihan**, Antara Mahmud, Antonios Anastasopoulos, Marcos Zampieri. “OffMix-3L: A Novel Code-Mixed Dataset in Bangla-English-Hindi for Offensive Language Identification.” *1st Workshop in South East Asian Language Processing (SEALP @ AACL 2023)*, Bali, Indonesia. [PDF] [Dataset]
- W1** **Md Nishat Raihan**, Dhiman Goswami, Antara Mahmud, Antonios Anastasopoulos, Marcos Zampieri. “SentMix-3L: A Bangla-English-Hindi Code-Mixed Dataset for Sentiment Analysis.” *11th International Workshop on Natural Language Processing for Social Media (AACL 2023)*, Bali, Indonesia. [PDF] [Dataset]

Preprints and Technical Reports

- P6** **Nishat Raihan**, Christian Newman, Marcos Zampieri. “A Taxonomy of Programming Languages for Code Generation.” *arXiv:2604.00239*, 2026. [PDF]
- P5** Mohammed Latif Siddiq, Noshin Ulfat, **Nishat Raihan**, Joanna C. S. Santos, Marcos Zampieri. “Multi-SaLLM: A Multilingual Security Assessment of Generated Code.” *Preprint*, 2025. [PDF]

- P4** **Md Nishat Raihan**, Dhiman Goswami, Antara Mahmud. “Mixed-Distil-BERT: Code-mixed Language Modeling for Bangla, English, and Hindi.” *arXiv:2309.10272*, 2023. [PDF] [Model]
- P3** **Md Nishat Raihan**. “Determining the Optimal Number of Clusters for Time Series Datasets with Symbolic Pattern Forest.” *arXiv:2310.00820*, 2023. [PDF]
- P2** R. Kushol, **M. Raihan**, M. S. Salekin, A. B. M. Rahman. “Contrast Enhancement of Medical X-Ray Image Using Morphological Operators with Optimal Structuring Element.” *arXiv:1905.08545*, 2019. [PDF]
- P1** **Md Raihan**, Hossain Mohammad Seym. “An Effective Navigation System Combining both Object Detection and Obstacle Detection Based on Depth Information for the Visually Impaired.” *B.Sc. Thesis, Dept. of Computer Science and Engineering*, 2018. [PDF]

RESEARCH ARTIFACTS: DATASETS AND MODELS

Full collection: [HuggingFace \(md-nishat-008\)](#) • [GitHub \(mrailhan-gmu\)](#)

Datasets

- **mHumanEval-Benchmark**: 836,400 coding prompts across 204 natural languages and 25 programming languages. [HuggingFace] [GitHub]
- **Bangla-Instruct**: 100K native Bangla instruction-response pairs (self-instruct, expert teacher models) for fine-tuning Bangla LLMs. [HuggingFace]
- **Bangla-TextBook**: 10M-token corpus from 163 national-curriculum textbooks (Grades 6–12) for LLM pretraining. [HuggingFace]
- **Mojo-Corpus**, **Mojo-SFT**, **Mojo-mSFT**: a 6.5M-token pretraining corpus and monolingual/multilingual instruction sets for the Mojo language. [HuggingFace] [GitHub]
- **CSEPrompts**: 200+ coding prompts and 100+ MCQ prompts for introductory CS evaluation. [GitHub]
- **MentalHelp**: 14M semi-supervised mental-disorder detection records. [GitHub]
- **TB-OLID**: offensive language identification in transliterated and code-mixed Bangla. [GitHub]
- **SentMix-3L**, **OffMix-3L**: 1K human-generated Bangla-English-Hindi code-mixed datasets for sentiment and offensive language. [HuggingFace] [GitHub]
- **Code-Mixed Sentiment** and **Code-Mixed Offensive** (Bangla-English-Hindi): 100K synthetic code-mixed datasets each. [GitHub]

Models

- **TigerLLM family**: state-of-the-art Bangla LLMs, released as instruction-tuned **1B** and **9B** variants. [1B-it] [9B-it]
- **TigerCoder**: Bangla code-generation model built on the TigerLLM base. [GitHub]
- **Mojo-Coder family**: base (*Mojo-Coder*), instruction-tuned (*Mojo-Coder-it*), and multilingually instruction-tuned (*Mojo-Coder-it-m*) LLMs for the Mojo language. [HuggingFace]
- **Tri-Distil-BERT** and **Mixed-Distil-BERT**: encoders pretrained on Bangla-English-Hindi and 560K code-mixed records. [HuggingFace]

INVITED TALKS

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| “Adapting Code LLMs for Under-Explored Domains” (<i>invited talk</i>), <i>California State University, Fresno</i> | May 12, 2026 |
| “Adapting Code LLMs for Under-Explored Domains” (<i>invited talk</i>), <i>Prairie View A&M University</i> | May 6, 2026 |
| “Adapting Code LLMs for Under-Explored Domains” (<i>invited talk</i>), <i>Duke University</i> | Apr. 21, 2026 |

EXPERIENCE

Research

Provost's Postdoctoral Fellow

University of Notre Dame, Dept. of Computer Science & Engineering

Aug. 2026 – 2028

Notre Dame, IN, USA

- Research on safety-by-construction guardrails and multilingual program synthesis for Code LLMs, mentored by Prof. Joanna C. S. Santos.

Graduate Research Assistant

George Mason University, Dept. of Computer Science

May 2024 – present

Fairfax, VA, USA

- Lead end-to-end LLM development: data curation, pretraining, fine-tuning, and evaluation for code and multilingual settings.
- Author and co-author peer-reviewed papers; present findings at international venues; collaborate across NLP and software engineering teams.

Teaching

Graduate Teaching Assistant

George Mason University, Dept. of Computer Science

Aug. 2021 – May 2024

Fairfax, VA, USA

- Supported four courses over eight semesters: CS 222 (Computer Programming for Engineers), CS 262 (Introduction to Low-Level Programming), and CS 583 (Analysis of Algorithms).

Lecturer, Computer Science

Uttara University

Jan. 2019 – Apr. 2019

Dhaka, Bangladesh

- Taught theory and lab courses (Introduction to Computers, C, C++); participated in computer vision research and student mentoring.

Industry

Software Engineer, RPA & OCR Development

Samsung R&D Institute Bangladesh

Nov. 2019 – Aug. 2021

Dhaka, Bangladesh

- Built production-grade RPA bots and services (Automation Anywhere, UiPath) for international organizations; integrated OCR engines (Tesseract 4, Google Vision, ABBYY, Azure, Tegaki) into IQ Bots with custom Python pipelines. Platforms: Automation Anywhere, UiPath, OpenText, SAP.

TEACHING

Graduate Teaching Assistant, George Mason University (8 semesters, 4 courses)

Analysis of Algorithms (CS 583, graduate)

- Led labs for a required MS/Ph.D. course of 75–90 students. Designed a “buggy-algorithm” lab method in which students diagnose a subtle partitioning bug in a flawed Quicksort, construct a test case that reliably exposes it, and defend their fix, turning assessment into authentic inquiry.

Introduction to Low-Level Programming (CS 262, undergraduate)

- Supported one of the department's most demanding courses (12 labs, 3 projects, midterm, final). Held extended office hours and introduced AI Code Review exercises in which students critique partially working ChatGPT output to find the breaking edge case.

Computer Programming for Engineers / Introduction to Programming (CS 222)

- Taught beginners from varied backgrounds; built a low-stakes environment for recursion by sharing and dissecting my own early buggy attempts to normalize the learning curve.

Course Development (proposed)

- *Responsible AI Deployment*: an upper-level elective bridging foundational algorithms and AI-augmented practice, covering LLM-assisted code generation, debugging, and the ethics of responsible deployment.

MENTORING

- **Jayoti Devi** (Ph.D. student, first year): mentorship yielded an ARR submission on AI guardrail and a comprehensive literature review; co-author on CodeGuard (EACL 2026).
- **Sadiya Sayara Chowdhury Puspo** (Ph.D. student, first year at the time): guidance produced *MentalHelp*, the largest multitask mental-health dataset of its kind, published at LREC-COLING 2024.

PROFESSIONAL SERVICE

Organizing Committee

- Organizer, BLP-2025 Shared Task on Code Generation in Bangla, the first shared task in this area, which attracted 158 international participants.
- Organizing Committee, 1st Workshop on Language Models for Low-Resource Languages (LoResLM @ COLING 2025).
- Organizing Committee, 1st and 2nd Workshops on Bangla Language Processing (BLP @ EMNLP 2023; BLP @ AACL 2025).
- Organizing Committee, Multilingual Lexical Simplification Pipeline (MLSP) Shared Task 2024.
- Organizing Committee, 3rd International Conference on Foundation and Large Language Models (AACL 2025).

Reviewing

- ACL Rolling Review (ARR) 2024 and 2025 (every cycle); LREC 2026; SIGCSE 2024–2026; COLING 2024–2025; SemEval 2024.

FUNDING EXPERIENCE

- Research supported in part by the Naval Sea Systems Command (NAVSEA) and multiple STTR awards, with hands-on exposure to proposal development, budget planning, and articulating measurable research impact.

LEADERSHIP AND VOLUNTEER ACTIVITIES

Member <i>The National Society of Leadership and Success (NSLS), George Mason Chapter</i>	Oct. 2024 – present
Vice-President <i>Bangladesh Graduate Student Association (BDGSA), George Mason University</i>	Apr. 2023 – Mar. 2024
Director of Sports <i>Bangladesh Graduate Student Association (BDGSA), George Mason University</i>	Apr. 2022 – Mar. 2023

REFERENCES

Attached separately.